Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 15)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

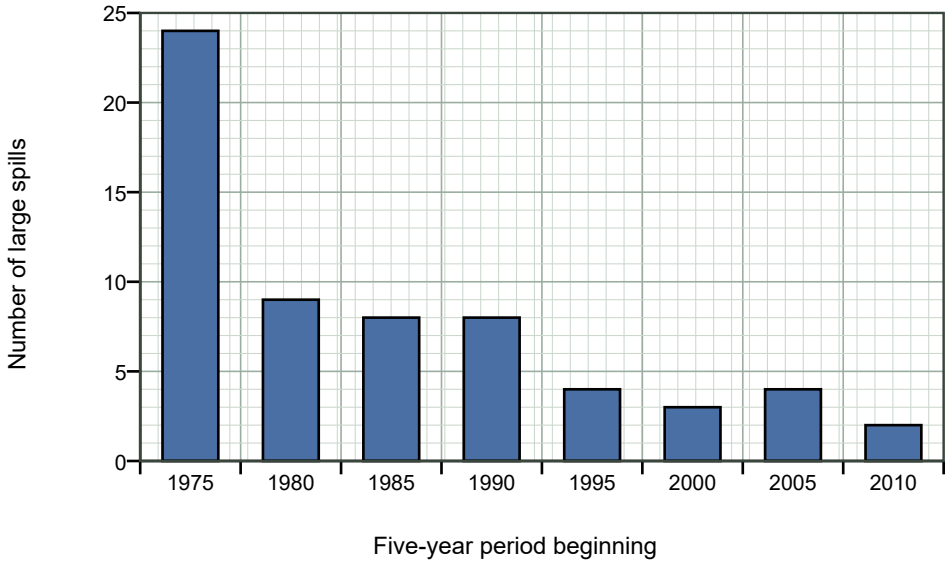


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 1 . 1 Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

Answer ____ %

0 1 . 2 Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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0 1 . 3 Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

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0 1 . 4 Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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0 2 . 1 Explain why organic pollution of rivers is usually monitored by measuring Biochemical Oxygen Demand (BOD) rather than the concentrations of the pollutants themselves.

[3 marks]

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0 2 . 2 Explain why a coliform count is used to assess the health risk from sewage contamination rather than testing for pathogens such as typhoid.

[2 marks]

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0 2 . 3 Give **three** characteristics of species that make them useful in a biotic index.

[3 marks]

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0 2 . 4 Give **one** advantage and **one** disadvantage of using a biotic index rather than chemical monitoring.

[2 marks]

Advantage .....

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Disadvantage .....

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0 3 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

0 3 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 3 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 3 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 3 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 3 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 4 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 4 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 5 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 5 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

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0 5 . 2 Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 5 . 3 Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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0 6 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 6 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

0 6 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

0 7 . 1 Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

0 7 . 2 Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

0 7 . 3 Describe **two** ways captured CO_2 can be stored and give **one** concern about long-term storage.

[3 marks]

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0 8 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 8 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 8 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 8 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 9 An aquifer beneath an agricultural plain holds 3.6 x 10^{11} m^3 of water. It is recharged at 1.2 x 10^9 m^3 per year, and 4.8 x 10^9 m^3 per year is abstracted from it.

0 9 . 1 Residence time = volume of the reservoir / rate of flow through it. Calculate the residence time of the water in this aquifer, using the recharge rate.

[2 marks]

Answer _____ years

0 9 . 2 Explain the properties a rock must have to form a useful aquifer.

[3 marks]

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0 9 . 3 Explain why the water table beneath the plain is falling.

[2 marks]

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0 9 . 4 Describe **four** consequences of continuing to over-abstract this aquifer.

[4 marks]

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0 9 . 5 Explain why an aquifer holding 'fossil water' cannot be exploited sustainably, and describe **two** measures that would reduce abstraction.

[4 marks]

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1 0 . 1 Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

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Extra space

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1 0 . 2 State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS